

Suricata NIDS Configuration Guide

Purpose

Suricata operates as the Network Intrusion Detection System within the VSOC environment. It is deployed directly on the pfSense firewall to ensure total visibility of:

- Ingress traffic entering from the transit network
- Egress traffic leaving internal hosts
- Lateral movement between internal hosts
- Attacker → victim traffic paths

This placement guarantees that all meaningful network activity is inspected and logged.

1. Deployment Model

Suricata is installed as a **pfSense package** and bound to the **LAN interface**, which carries all internal SOC traffic.

Rationale

- pfSense sees all routed packets between attacker, victims, SIEM, and WAN.
 - Avoids complexity of a separate NIDS VM (no port mirroring required).
 - Simplifies rule updates and log forwarding.
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2. Installation Steps

Step 1 — Install Suricata

pfSense GUI:

System → Package Manager → Available Packages

Search "Suricata" → Install

Wait for automatic dependency installation.

3. Interface Configuration

Step 1 — Add Interface

Services → Suricata → Interfaces → + Add

Select: - **Interface:** LAN - **IPS Mode:** Disabled (initial testing only) - **Block Offenders:** Unchecked

IPS mode can be enabled later after tuning.

Step 2 — Save and enable the interface.

4. Ruleset Configuration

Enabling Emerging Threats Open Rules (ET Open)

Services → Suricata → Global Settings
Check: Enable ET Open Rules
Save

Selecting Rule Categories (LAN interface)

Recommended categories: - emerging-scan.rules - emerging-malware.rules - emerging-web.rules - emerging-trojan.rules - emerging-policy.rules

Force Rules Update

Services → Suricata → Updates → Force Rules Update

If rules fail to download: - Verify WAN gateway - Test external DNS: `ping rules.emergingthreats.net` in pfSense shell - Ensure pfSense has outbound Internet connectivity

5. Starting Suricata

Services → Suricata → Interfaces → Start (►)

Verify service in:

Status → Services → Suricata

6. Log Locations

Suricata stores alerts and event logs in:

```
/var/log/suricata/fast.log      # Summary alerts
/var/log/suricata/eve.json     # Full structured event data
```

These logs feed directly into Wazuh via syslog-ng.

7. Validation Tests

Test 1 — Basic IDS Signature

From any internal host:

```
curl http://testmyids.com
```

Expected alert: ET POLICY "Test My IDS" signature.

Test 2 — Port Scan Detection

From Kali attacker → victim:

```
nmap -sS -T4 -p1-1000 <victim_ip>
```

Expected alerts: - ET SCAN Nmap Scripting Engine User-Agent Detected - ET SCAN Potential SSH Scan - ET SCAN Suspicious inbound connection

Important: scanning must **traverse pfSense**.

If both hosts are on the same OpenStack internal switch without routing, Suricata will not see packets. Ensure all traffic paths route **through pfSense LAN**.

8. Known Behavior and Troubleshooting

Symptom: No alerts generated

- Confirm interface is enabled

- Confirm traffic crosses pfSense
- Disable IPS mode for testing
- Ensure rule categories are enabled

Symptom: Rules fail to update

- Fix DNS: add DNS to transit and LAN subnets
- Ensure default gateway on WAN is correct

Symptom: False positives

- Tune by disabling noisy rules inside Suricata category lists
- Recommended: tune only after completing SIEM integration

9. Integration With Wazuh SIEM

Log forwarding is performed via **syslog-ng** because pfSense's default syslog daemon: - Cannot forward JSON logs correctly - Conflicts with Suricata's EVE JSON output

Syslog-ng pulls logs from Suricata:

```
source s_suricata_eve {
    file("/var/log/suricata/<interface-folder>/eve.json" flags(no-parse) follow-
freq(1));
};
```

Then sends them to Wazuh Manager:

```
destination d_wazuh { udp("<SIEM-IP>" port(514)); };
```

A full forwarding pipeline is described in the Syslog-ng Forwarding Guide.

10. Suricata's Role in the VSOC Environment

Suricata provides: - Packet-level monitoring - Threat signature detection - Behavioral analysis (scans, brute force, malware activity) - JSON logs for SIEM correlation - Real-time alerting for classroom/lab demos

Its placement on pfSense ensures visibility into every attack chain demonstrated in the VSOC.